

Project Write-up:
TOXICDDI EXPLORER:
VISUALIZING TOXIC DRUG-DRUG INTERACTIONS BY TYPE AND SEVERITY

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1 Introduction

1.1 Motivation and Significance

Drug-drug interactions are a well-known source of adverse drug reactions (ADRs), which contribute significantly to patient morbidity, hospitalizations, and even fatalities [1]. However, despite the availability of large DDI datasets, the information is often presented in static, text-based formats that are difficult to navigate and interpret, especially when considering multiple drugs, toxicity types, and interaction mechanisms [4]. Currently available drug interaction checkers primarily focus on general adverse effects or binary warnings (e.g., “avoid combination” or “monitor closely”) without providing a deeper breakdown of the specific toxicological outcomes or mechanistic pathways involved. Few existing tools offer interactive, visual exploration of toxic DDIs by toxicity category (e.g., liver, kidney, heart, nervous system) and severity. This lack of user-centered, visual interfaces poses a challenge for healthcare professionals, researchers, and students who need to understand and assess the risks of harmful drug combinations.

1.2 Project Objectives

This project addresses the core question: ***"How can we effectively visualize toxic drug-drug interactions (DDIs) to support safer prescribing practices and improve understanding of toxicity risks across different drug combinations?"***

To answer this, we designed and developed an interactive dashboard that visualizes toxic DDIs using network graphs, severity-distribution charts, and searchable interaction summaries. The data used in the project comes from curated sources such as DrugBank and the DDInter database, and includes interaction types, toxicity classifications, and severity annotations. The key objectives of the project are:

- To create an intuitive and interactive network visualization of toxic DDIs
- To allow users to explore interactions by drug, toxicity type, and severity level
- To provide detailed breakdowns of individual drugs' interaction profiles
- To support targeted queries of drug pairs for clinical reference or academic analysis

The final product is a Shiny dashboard that integrates multiple views to support exploratory analysis, education, and reference use for healthcare professionals, students, and researchers.

1.3 Novelty

Unlike most existing DDI tools, which emphasize static lists or one-dimensional alerts, this project introduces a dynamic and modular visualization platform with user interaction at its core. The dashboard incorporates several key innovations:

- **Interactive Network Graphs:** Visualizing DDIs as interconnected nodes and edges, with filters for toxicity type and severity.
- **Multi-View Exploration:** The application includes three distinct views: (1) a network-wide overview of all interactions, (2) a focused view of interactions for a single drug, and (3) a customizable drug pair query tool. Each view is tailored to a specific exploration task.
- **Dynamic Filtering and Customization:** Users can filter by severity, toxicity type, or drug of interest; view interaction descriptions; and search interactively, enhancing usability and insight generation.

This project bridges the gap between large-scale biomedical data and actionable insights. By enabling interactive, visual exploration of DDIs, the tool makes toxic interaction data more accessible, interpretable, and useful across multiple domains.

The code for our project is available at <https://github.com/chipphan1110/toxic-ddi-visualization.git>

2 Our Approach

2.1 Data Preparation and Preprocessing

To construct a reliable dataset for toxic drug-drug interaction (DDI) visualization, we implemented a two-stage pipeline: *data preparation* and *data processing*. The preparation stage focuses on collecting and enriching raw data, while the processing stage transforms the data into a clean, structured format optimized for interactive analysis.

2.1.1 Data Preparation

This stage involved collecting raw interaction records, identifying toxicity-related cases, and enriching them with clinically relevant information from external sources.

- **Initial Data Collection:** We obtained the initial dataset from Mendeley Data [6], which included drug interaction data sourced from DrugBank [3]. This consisted of three files: one with raw DDI records, one mapping raw interaction types to merged toxicity categories, and one summarizing merged interaction types and their frequencies.
- **Data Normalization:** We curated a list of ten toxicity types, including cardiotoxicity, nephrotoxicity, hepatotoxicity, neurotoxicity, and others. The `interaction_type` column in the raw dataset was cleaned and matched against this list to identify toxicity-related interactions. Drug pairs that matched at least one toxicity keyword were retained, and a new column `matched_toxicity_types` was added to annotate the types of toxicity involved in each interaction.
- **Extra Crawling:** To enrich the dataset with clinical context, we implemented an automated web crawling pipeline using Selenium. For each drug pair, our crawler accessed the DDInter interaction checker database [5] and extracted the severity level of the interaction, a detailed clinical description, and recommended management guidance. These additional attributes were merged into the dataset as new columns: `ddinter_severity`, `ddinter_interaction`, and `ddinter_management`.

The result of this preparation stage is the `data/ddi_with_ddinter_info.csv` file, which contains 2,144 unique drug-drug interactions identified as toxic. The dataset includes 10 columns: drug identifiers and names, normalized interaction types, matched toxicity categories, and enriched clinical annotations from the DDInter database. This data serves as the foundational input for downstream processing and visualization.

2.1.2 Data Processing

This stage focused on transforming the enriched dataset into a structured, visualization-ready format by applying cleaning steps, visual mappings, and network structuring techniques suitable for interactive exploration.

- **Initial Cleaning:** We loaded the prepared dataset from `data/ddi_with_ddinter_info.csv`, removed rows with missing or incomplete information, and eliminated duplicate drug pairs. Drug identifiers and names were also standardized to ensure consistency across all entries.
- **Visual Mapping Definition:** A color scheme was established to represent different toxicity types (e.g., red for cardiotoxicity, blue for nephrotoxicity), and a severity-based edge weight system was introduced to reflect interaction risk (e.g., major: 10, moderate: 8, minor: 5). Unknown or unspecified values were mapped to default colors and minimal weights to maintain visual coherence.
- **Edge List Construction:** We created a list of edges representing toxic interactions between drug pairs. Each edge includes source and target drug IDs, toxicity category, severity level, DDInter interaction description, and corresponding visual properties such as edge color and thickness. This enables encoding of both interaction type and risk severity.
- **Node List Construction:** Each drug was analyzed to determine the dominant toxicity type across its interactions. A color was assigned based on this dominant type, and a node list was compiled including drug IDs, names, and visual attributes used in the network visualization.
- **Network Structuring:** The edge and node data were merged into a unified network format. Consistency checks were applied to align color encoding and ensure all entities were represented. Visual enhancements, such as tooltips and fallback styling, were included to improve interactivity and clarity.

The complete processed network was saved as `data/network_data.rds`, containing all necessary metadata, including edges, nodes, and styling information, for integration into the Shiny dashboard. This processing pipeline transforms raw enriched DDI records into a structured and interactive-ready format. By encoding toxicity types as colors,

severity as edge thickness, and dominant drug effects as node attributes, the resulting network data enables intuitive and clinically meaningful exploration of toxic interactions.

2.2 Data Visualization Techniques

2.2.1 Design Rationale

The visual design of this dashboard is driven by the analytical needs of users exploring toxic drug-drug interactions. Our goal was to develop a coherent set of visualizations that support different levels of inquiry: from high-level patterns in the global interaction network to detailed inspection of individual drug effects. To accomplish this, we adopted a modular layout, separating the interface into three distinct but semantically linked views. Each view was guided by established principles in data visualization, including the *Grammar of Graphics*, perceptual effectiveness, and interactive design patterns for exploratory analysis.

1. Network Overview: Visualizing Global Toxicity Structure

The Network Overview tab is designed to provide an overall picture of the toxic DDI landscape. It allows users to identify central drugs, discover clusters of interactions with similar toxicity types, and assess severity patterns at the network level. A network diagram was selected because it effectively conveys relational structure and connectivity. Each drug is represented as a node and each toxic interaction as an edge. Visual encodings are applied as follows:

- **Edge thickness** encodes severity level, mapping quantitative information to line width, a perceptually effective channel.
- **Edge color** represents toxicity type using distinct hues for categorical distinction.
- **Node size** corresponds to degree centrality, highlighting highly interactive (hub) drugs.
- **Node color** reflects the dominant toxicity type among a drug's interactions.

This design supports the task of pattern discovery and structural comparison. It aligns with Shneiderman's visual information-seeking mantra: *"Overview first, zoom and filter, then details on demand"* [2]. Interactivity, such as zooming, dragging, and tooltips, further enhances exploration.

2. Single Drug View: Focused Analysis of Drug-Centric Interactions

The Single Drug View is intended for users interested in exploring the interaction profile of a specific drug. It answers questions such as: *"What are the most severe toxicities associated with this drug?"* and *"Which other drugs are involved in those interactions?"*. This tab combines two complementary visualizations:

- **A localized interaction network**, centered on the selected drug and showing all directly connected drugs. Interaction edges are styled consistently with the global network: color for toxicity type, thickness for severity, and tooltips for interaction descriptions. The focal drug is visually highlighted to support user orientation.
- **A sunburst chart** showing the distribution of interaction severities by toxicity type. This radial visualization uses a hierarchical layout with two levels: interaction type (inner ring) and severity category (outer ring). The sunburst was selected to communicate part-to-whole relationships and breakdowns, important for understanding the composition of risks.

This dual-view interface supports both topological (who is involved) and compositional (how severe and why) reasoning, aligning with multivariate exploration principles in dashboard design.

3. Drug Pair Query: Direct Lookup and Clinical Interpretation

The Drug Pair Query tab provides a simple, text-based interface for users to explore specific combinations of interest. When users select 2 to 5 drugs, the system returns all toxic interactions among them in a structured format. A **card-based layout** was chosen to organize the interaction details clearly and readably. Each card displays:

- Drug pair names and IDs
- Toxicity type and severity level
- A brief clinical interaction description and management information, or a fallback "N/A" for missing data

Visual emphasis is added through color-coded severity tags (e.g., red for Major, gray for Unknown), enabling rapid visual scanning and comparison. This view supports lookup tasks and clinical decision-making where visual structure is less important than clarity and completeness.

2.2.2 Choice of Libraries and Tools

To implement the dashboard's interactive and visually consistent interface, we utilized a combination of R libraries and Shiny extensions. The selection of tools was based on their flexibility, integration with Shiny, and support for interactivity and customization. The following summarizes the key libraries used and their roles in the application:

- **visNetwork:** Used as the primary engine for network visualization. It offers extensive control over node and edge styling, layout physics (via `vis.js`), and interactivity features such as tooltips, zooming, dragging, and dynamic filtering. Its ability to bind R data frames directly to visual elements allowed for seamless integration of node and edge attributes.
- **plotly:** Employed to create the sunburst chart in the Single Drug View. Plotly's support for hierarchical visualizations, custom hover templates, and smooth interactivity made it ideal for depicting nested part-to-whole relationships in severity distribution.
- **dplyr and tidyr:** Utilized extensively for data wrangling tasks, including filtering, grouping, joining, and reshaping data. These tidyverse libraries enabled efficient and readable transformations required for reactive server logic and visualization preparation.
- **scales:** Used to normalize quantitative variables (e.g., node degrees) into visual scales such as node size or opacity. The `rescale()` function allowed for dynamic visual mappings while maintaining interpretability.
- **shiny.fluent and shiny.react:** These libraries provided the UI components and layout framework based on Microsoft's Fluent Design System. They enabled the creation of modern and responsive UI elements such as dropdown selectors, cards, stacks, and input panels, with a consistent and accessible visual style.

3 Final Product Description

3.1 Overview of the Dashboard

The *Toxic Drug-Drug Interaction Dashboard* is an interactive web application developed using the R **Shiny** framework. It enables the exploration of toxic drug-drug interaction (DDI) patterns through a dynamic and modular user interface. Designed primarily for research and reference purposes, the dashboard allows users to visualize and query known toxic interactions across various drug combinations.

The dashboard is powered by several modern R packages, including **shiny**, **shiny.fluent**, **shinycssloaders**, **visNetwork**, and **shinyjs**. The application is structured using modular components, with separate files managing the user interface and server logic for each core view. Data is sourced from a pre-processed RDS file, **network_data.rds**, which contains the full set of interaction nodes and edges.

Upon launching the application, users are greeted with a well-structured landing page comprising the following components:

- **Header Bar:** Contains the application title and a concise description to establish context.
- **Sidebar Navigation:** Provides access to the three main views: *Network Overview*, *Single Drug View*, and *Drug Pair Query*.
- **Disclaimer Panel:** Presents a cautionary note regarding the limitations of the dataset and the intended use of the results.
- **Main Display Area:** Dynamically renders tab-specific content. In the default view, an interactive network graph is shown, visualizing known toxic interactions between drugs.

The primary visualization, as seen in Figure 1, features a force-directed network layout where nodes represent drugs and edges denote interactions. Different colors are used to distinguish between interaction clusters, and interactivity allows users to explore local subgraphs, filter by interaction type or severity, and identify key hubs in the network.

This modular and visually driven design ensures that the dashboard supports both exploratory analysis and targeted hypothesis testing, making it a useful tool for data scientists, pharmacologists, and researchers interested in drug safety.

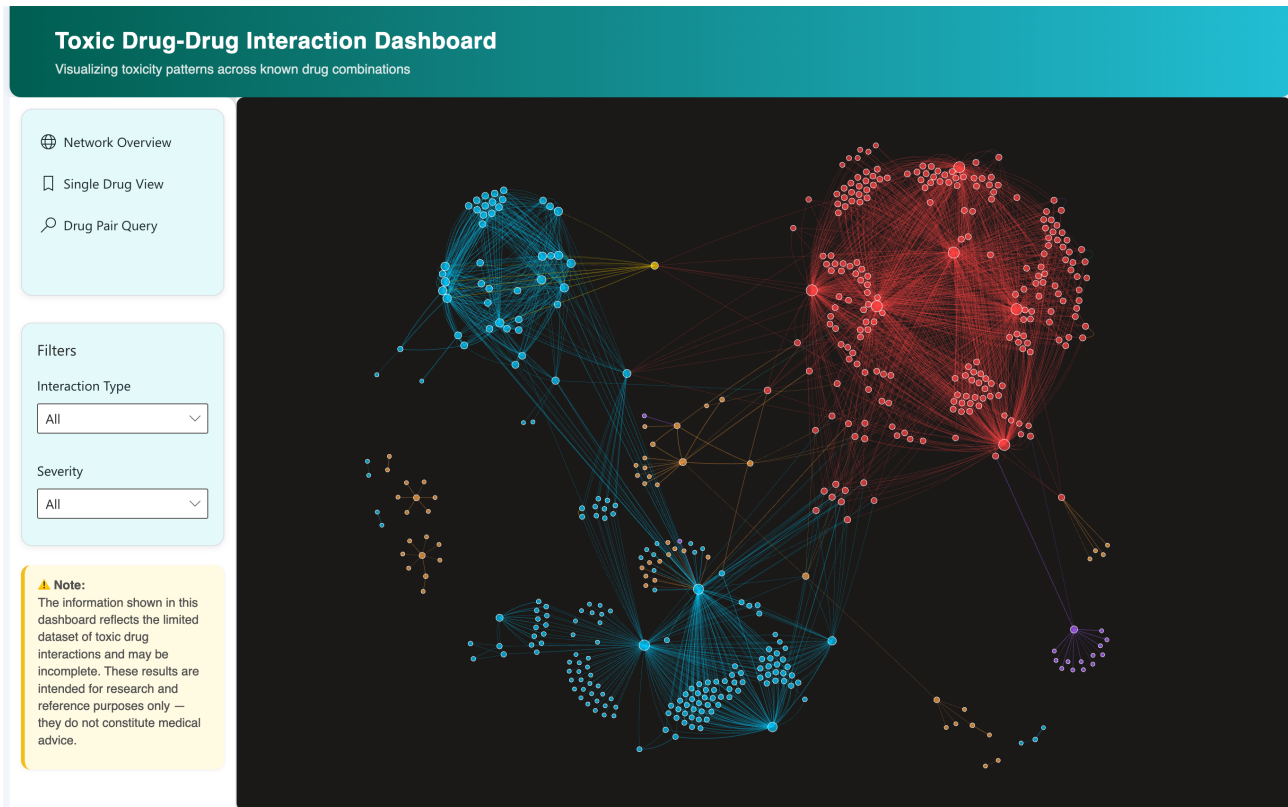


Figure 1: Main dashboard view with network visualization, sidebar navigation, and disclaimer.

3.2 Tabs and Features

3.2.1 Network Overview Tab

The *Network Overview* tab functions as the central visualization interface within the Toxic Drug-Drug Interaction Dashboard. It provides a comprehensive, interactive graph-based representation of known toxic interactions between pharmaceutical compounds. Each node in the network corresponds to a unique drug entity, and edges between nodes indicate documented toxic interactions, with visual clustering often suggesting shared pharmacological or mechanistic properties among the drugs.

Upon selecting the *Network Overview* from the left-hand navigation panel, the main interface dynamically loads a network visualization alongside a sidebar panel that contains filtering controls. This sidebar introduces two key filtering options:

- **Interaction Type:** This dropdown allows users to specify the toxicity mechanism or organ system affected by the interaction. Examples of available interaction types include **nephrotoxicity**, **hepatotoxicity**, and **neurotoxicity**. Selecting a particular interaction type filters the network to show only those drug pairs known to induce the specified toxicity.
- **Severity:** This control enables users to refine the visualization further based on the clinical severity of the interaction. Available severity levels may include **Major**, **Moderate**, **Minor**, and **Unknown**. Alternatively, users can opt to display all severity levels by selecting **All**.

These filters are designed to work in conjunction, allowing users to investigate specific subsets of the toxic interaction network based on both type and severity. When either or both filters are applied, the network graph updates in real time to reflect only the relevant nodes and edges corresponding to the selected criteria.

If no interactions are found that match the current filtering conditions, the dashboard provides clear and immediate feedback to the user through a notification box. This message, shown in the main panel, reads:

No interactions match the selected filters.

This ensures transparency and prevents user confusion when the dataset lacks matches for a particular query.

Figure 2 is an example illustrates the dashboard after selecting the **nephrotoxicity** interaction type and choosing **All**

severities. The resulting visualization displays a network consisting solely of drug pairs associated with nephrotoxic interactions. Nodes are sized based on the number of interactions (degree centrality), and edges represent the reported toxic links. This focused view helps researchers and clinicians quickly identify key drugs that may pose nephrotoxic risks when used in combination.

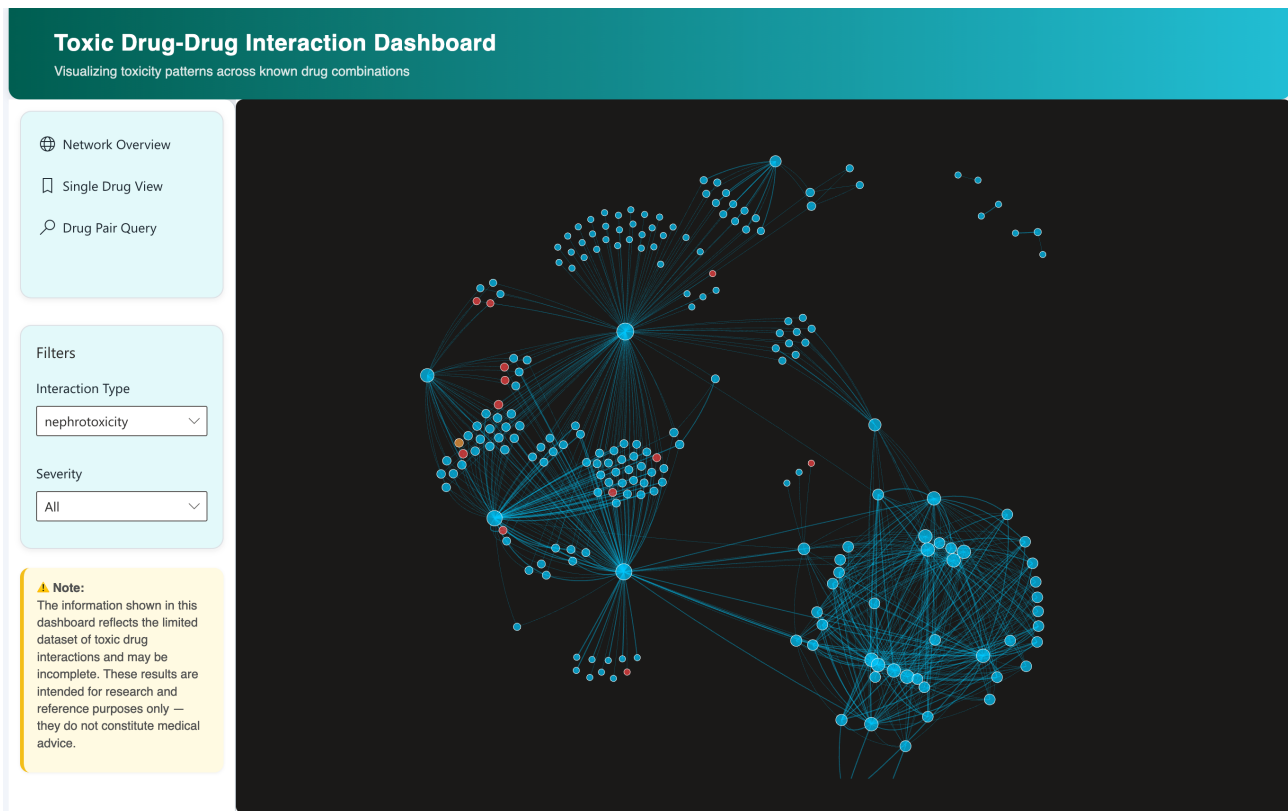


Figure 2: Filtered network showing nephrotoxic drug interactions with active filters applied.

3.2.2 Single Drug View Tab

The **Single Drug View** tab provides an intuitive interface for examining all known toxic interactions associated with a particular drug. Users begin by selecting a drug from the dropdown menu labeled **Drug Selection** in the left-hand sidebar. An informative disclaimer below the dropdown warns users that the visualizations are based on a limited dataset and are intended for research purposes only.

Once a drug, such as **Acetyldigitoxin**, is selected, two visual components are rendered on the main panel (see Figure 3):

1. **Drug Interaction Network:** This interactive network graph centers the selected drug and displays all other drugs that have reported toxic interactions with it. In the example of **Acetyldigitoxin**, the graph reveals a dense network of 224 interacting drugs, each connected by red edges that symbolize known toxic links. Hovering over a node provides further detail, such as the name of the interacting drug.
2. **Interaction Severity Distribution:** Below the network graph is a concentric ring visualization showing a summary of interaction severity and toxicity type. For **Acetyldigitoxin**, the plot reveals that all toxic interactions are classified as **minor** in severity and fall under the category of **cardiotoxicity**. This plot is generated using **plotly** and includes interactive features such as hoverable labels and export options.

3.2.3 Drug Pair Query Tab

The **Drug Pair Query** tab allows users to investigate potential toxic interactions among two to five specific drugs. The user selects the desired drugs using a multi-select dropdown input located at the top of the panel. Once the selection is made, the dashboard generates a series of structured interaction cards, each displaying relevant details for a known interaction between a drug pair within the selection.

Each interaction card includes:

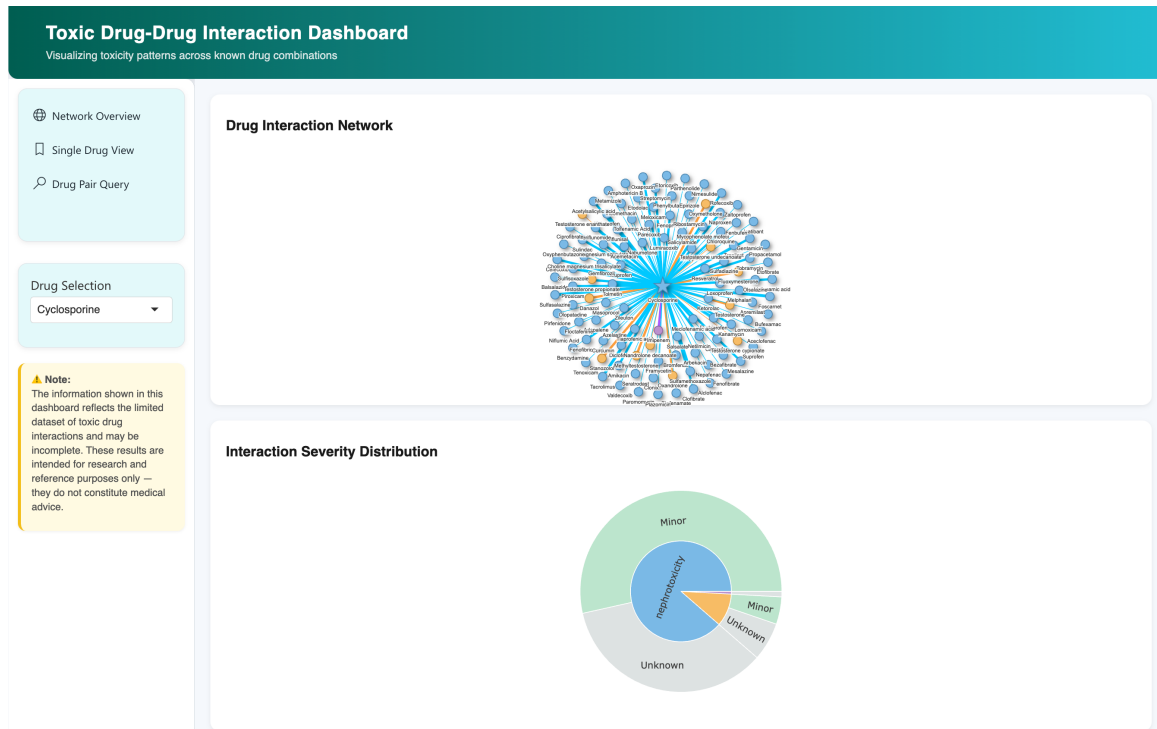


Figure 3: Single Drug View for Acetyldigitoxin, showing interaction network and severity distribution.

- **Drug Names and IDs** for both interacting agents
- **Interaction Type**, such as Nephrotoxicity
- **Severity Level**, classified into categories like Moderate or Major
- **Descriptive Summary** of the toxic mechanism or interaction context

For example, figure 4 demonstrates a case where three nephrotoxic antibiotics, **Bacitracin**, **Kanamycin**, and **Colistimethate**, have been selected. The dashboard identifies three toxic interactions among these drugs:

1. **Bacitracin** and **Colistimethate** exhibit a **Moderate** nephrotoxic interaction, potentially exacerbated by co-administration with glycopeptide or polypeptide antibiotics.
2. **Bacitracin** and **Kanamycin** have a **Major** interaction, likely due to additive or synergistic effects on renal function and neurological systems.
3. **Colistimethate** and **Kanamycin** also show a **Major** nephrotoxic risk.

If no known interaction exists between the selected drugs, the dashboard will return a message stating:

No interaction found between selected drugs.

This ensures transparency by explicitly confirming the absence of documented toxic interactions in such cases.

3.3 User Manual

3.3.1 System Requirements

- **R version:** 4.0.0 or newer
- **Required R packages:** shiny, shiny.fluent, shinycssloaders, visNetwork, shinyjs, plotly
- **Optional tools:** RStudio or Visual Studio Code with R extension

3.3.2 Launching the Application

1. **Download or Clone the Project:** Download or clone the GitHub repository and place all files in a single directory. Ensure that `app.R` and `data/network_data.rds` are present.

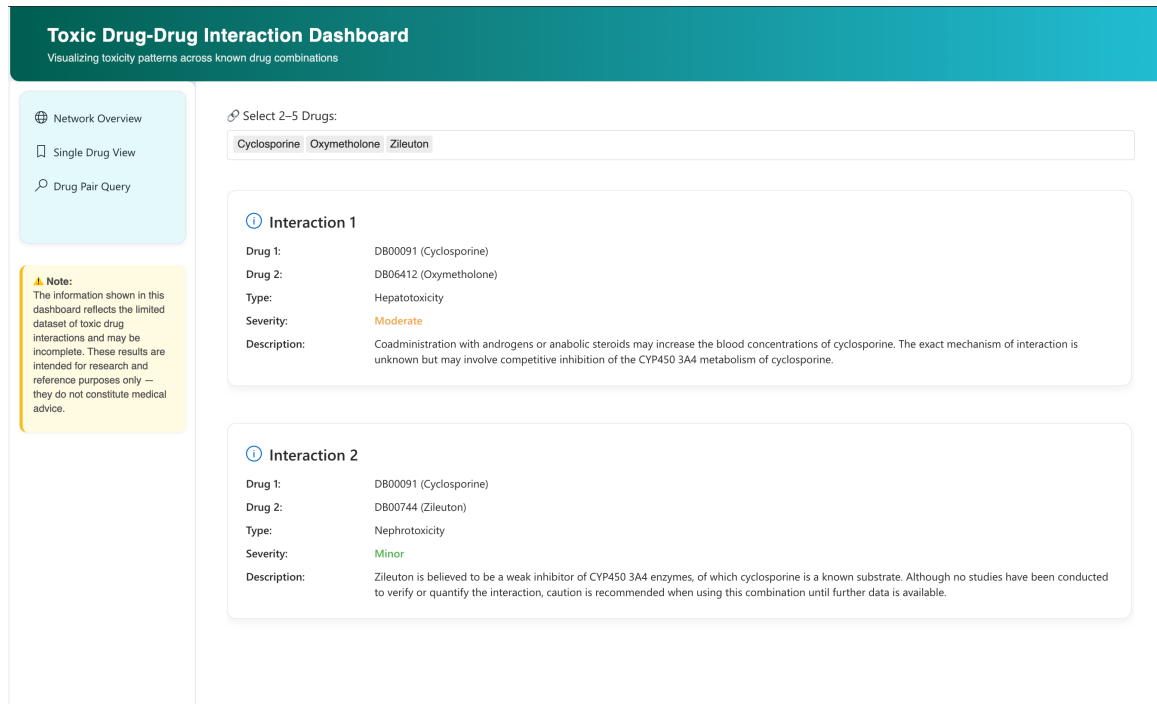


Figure 4: Drug Pair Query showing nephrotoxic interactions among Bacitracin, Kanamycin, and Colistimethate.

2. **Install Required Packages:** Open R or RStudio and run the following command:

```
install.packages(c("shiny", "shiny.fluent", "shiny.cssloaders",  
                  "visNetwork", "shinyjs", "plotly"))
```

3. **Run the Application:** In your R console or RStudio, set the working directory to the project folder and execute:

```
shiny::runApp()
```

4. **Access the Dashboard:** Once launched, the dashboard will open in your default web browser at a local address such as `http://127.0.0.1:xxxx`.

3.3.3 Using the Dashboard

Navigation: Use the left sidebar to switch between the three main tabs:

- **Network Overview:** Explore the global structure of toxic DDIs.
- **Single Drug View:** Analyze the interaction profile of a selected drug.
- **Drug Pair Query:** Investigate toxic interactions among a custom group of drugs.

Filters and Controls:

- **Network Overview Filters:**
 - *Interaction Type:* Filter interactions by toxicity category.
 - *Severity:* Filter interactions by severity level (Minor, Moderate, Major).
- **Single Drug Selection:** Use the searchable dropdown to select a drug. The top panel displays its interaction network, while the bottom panel shows severity distribution by toxicity type.
- **Drug Pair Query:** Select between two and five drugs from the dropdown to display all known interactions among the selected pairs.

3.3.4 Interpreting Results

Network Graphs: Nodes represent drugs, and edges represent toxic interactions. Edge color encodes toxicity type, and edge thickness reflects severity. Node size is based on interaction degree.

Details and Tooltips: Hover over or click on nodes and edges to display additional information such as drug names, interaction categories, severity levels, and clinical descriptions.

4 Discussion

4.0.1 Data Insights from the Visualization of Toxic Drug-Drug Interactions

The dataset includes a total of 503 unique drugs and 2,143 known toxic drug-drug interactions, representing a rich and clinically meaningful network. Visualization of this data reveals important structural and semantic patterns that would be difficult to grasp from tabular views alone.

The interaction network shows a highly imbalanced distribution of toxicity types. The vast majority of interactions are categorized as cardiotoxicity (60%) and nephrotoxicity (35%), which together account for more than 95% of all recorded interactions. This suggests that cardiovascular and renal effects are either the most frequently occurring or most studied forms of drug toxicity in current pharmacological datasets. In contrast, other toxicity types, such as hepatotoxicity (67 interactions) and neurotoxicity (19 interactions), are far less common, while ototoxicity and pulmonary toxicity are exceptionally rare. Notably, there is only one recorded case of pulmonary toxicity in the entire dataset. These discrepancies may reflect either true biological prevalence or data reporting biases. Drugs contributing to the major toxicity clusters form dense, highly connected communities, whereas the less frequent toxicity types appear as isolated nodes or small peripheral components. This imbalance highlights potential gaps in toxicity reporting and underscores the importance of more inclusive pharmacovigilance practices.

Moreover, the visualization reveals several high-degree nodes, drugs that are involved in a disproportionately large number of toxic interactions. These "hub drugs" may warrant closer investigation in clinical practice due to their potential to contribute to compounded toxicity in polypharmacy settings. Their prominence in the network aids in hypothesis generation and clinical prioritization.

4.1 Effectiveness of the Visualization Techniques

The dashboard's design employs a diverse set of visualization techniques, each optimized for a specific user task: structural exploration, focused drug profiling, or interaction lookup. Across all views, visual encodings are chosen to match the data type and perceptual effectiveness of the variable.

In the network view, toxicity type is encoded via distinct hues and severity via edge width, two high-ranking channels for encoding categorical and ordinal data, respectively. This allows users to distinguish clusters of similar toxicity and identify high-severity interactions with ease. The use of node size to represent degree centrality further enhances interpretability by spotlighting highly interactive drugs.

The sunburst chart in the Single Drug View provides a compositional snapshot of interaction types and severity distribution. By nesting severity within toxicity categories, this chart supports part-to-whole reasoning and allows users to visually assess the dominant toxicological profile of any single drug. This design complements the local interaction network and reinforces consistent semantic mappings.

The card-based layout in the Drug Pair Query tab prioritizes clarity and interpretability. Rather than overwhelming users with graph complexity, it presents key interaction fields, such as severity, description, and type, in a structured, readable format. Visual enhancements like color-coded severity tags and fallback labels (e.g., "N/A" for missing descriptions) maintain consistency and trustworthiness, even when the data is sparse or incomplete.

Together, these visualization strategies not only improve usability but also facilitate deeper insight into toxic DDIs, supporting safe prescribing, exploratory analysis, and pharmacological education.

5 Limitations

While the dashboard successfully supports exploratory analysis of toxic drug-drug interactions (DDIs), several limitations should be acknowledged regarding both the data and the visualization framework.

First, the underlying dataset, although curated from reputable sources such as DrugBank and DDInter, may not be comprehensive. The recorded interactions reflect only known and reported toxicity cases. As a result, many potentially

relevant or emerging interactions may be missing, especially for rare toxicity types. For example, pulmonary toxicity is represented by only a single interaction in the dataset, limiting meaningful analysis or visualization for that category.

Second, the dataset is inherently imbalanced in its representation of toxicity types. Over 95% of interactions are classified as either cardiotoxicity or nephrotoxicity, while categories such as hepatotoxicity, neurotoxicity, and ototoxicity are underrepresented. This imbalance may bias interpretation and reduce the effectiveness of visual comparisons across toxicity types.

Third, severity labels and interaction descriptions sourced from DDInter are not always standardized or complete. Some entries lack clear severity annotations or mechanistic explanations. Although the dashboard handles such cases using fallback logic (e.g., displaying “Unknown” or “N/A”), the absence of uniform information may affect interpretability and decision-making accuracy.

From a technical standpoint, the network visualization—while effective for moderate-sized graphs—may become visually cluttered as more filters are disabled or larger subsets of the network are rendered. This could affect performance and user experience in future expansions involving much larger datasets.

Lastly, the dashboard is not intended to be a clinical decision support system. It serves purely as a visualization and educational tool. The information provided is for research and reference purposes only and should not be used as a substitute for professional medical judgment.

6 Future Directions

This project establishes a foundational tool for visualizing toxic drug-drug interactions; however, several extensions could enhance its scope and utility. One promising direction involves enhancing the dataset by incorporating additional sources, such as pharmacovigilance databases and real-world clinical records. This would improve coverage and address current imbalances in toxicity type representation.

The user experience could also be extended by introducing features such as cross-view highlighting, interactive filtering between charts, and the ability to annotate or save findings. These enhancements would support more advanced analytical workflows and personalized exploration.

Finally, adapting the tool into a web-hosted platform or integrating it with decision-support systems could extend its impact to healthcare professionals and educators, enabling real-time exploration and interpretation of drug interaction risks.

References

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